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Research Paper

Instantaneous interface freezing induced abnormal icing dynamics during droplet impact icing

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ABSTRACT

In polar regions and extreme low temperature conditions, the droplet impact icing dynamics and freezing behavior are changed by the fast freezing at the interface and heat transfer between the droplet and the substrate. In this study, the droplet impact icing hydrodynamics on substrate from −29 ℃ to −64 ℃ are characterized and studied. The actual maximum spreading diameter is redefined based on the elaborated observation and the spreading area is obtained to reflect the more real interfacial hydrodynamics. The maximum spreading diameter generally decreases with the increase in substrate supercooling. However, the spreading diameter will not decrease monotonously and even increases with further increase in substrate supercooling when substrate is cold enough. The main reason can be attributed to the existing compressed air and fast interfacial freezing, characterized by the splashing of frost at the interface, which may serve as a lubrication layer with positive effect on the droplet spreading. Regarding the bulk freezing time, the model revised in this work can predict the cases with good thermal contact between ice lamella and substrate by incorporating the Schwartz model to calculate the interfacial temperature. The discontinuous contact induced by instantaneous interface freezing and thermal–mechanical relaxation can prolong the freezing time greatly. The present study not only characterized the abnormal icing behaviors associated with quick interfacial freezing during the spreading stage of droplet impacting freezing, but also proposed a revised model to predict the freezing time of droplet impact icing with good contact. The reported phenomena can provide more insights into the instantaneous interface freezing and inspire the anti-icing method in polar regions or deep space with extreme low temperatures.

1. Introduction

Ice accretion on solid surfaces poses safety threat and efficiency reduction to equipment [1]. Droplet impact icing is a ubiquitous phenomena both in nature and some industrial processes [2]. Compared with sessile droplet freezing which can be induced either by heterogeneous nucleation or homogeneous nucleation [3–12], droplet impact icing induced by heterogeneous nucleation at impact interface involves more strong and capricious interactions among droplet (ice), air and impacted surfaces [13–21].

In this century, researchers have developed numerous functional icephobic surfaces to achieve the repelling of droplets [16,17], delay of nucleation [21,22], lower adhesion strength [23] and even self-removal [24] by designing superhydrophobic [25–28], slippery [22], photothermal [29,30], and magnetic surfaces [31] etc. The effects of wettability [32], micro-nano structure [33], frost [34], surface movement [35], surface shape [36,37], property of droplet [38] and icing environment [7,39] on the icephobic performance of the above surfaces are thoroughly and systematically investigated. Results show that the desired icephobic properties always fail with the increase in humidity and/or surface supercooling, demonstrating the still requirement for new strategies to further improve their environmental compatibility.

In recent two decades, droplet impact and icing dynamics have been being a hot topic in research. Bartolo et al. [40] observed that the retraction dynamics changed from a capillary-inertial regime to a capillary-viscous regime on hydrophobic surfaces without cooling and proposed a semi-quantitative model for the solid–liquid contact line dynamics. Laan et al. [41] demonstrated a universal solution to rescale the maximum spreading behavior of a droplet that impacts on a surface

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Nomenclature

Symbols

A_m	Maximum spreading area, mm^2
C_p	Specific heat, J/(kg·K)
D_0	Diameter of water droplet, mm
D_m	Maximum spreading diameter, $\sqrt{4A_m/\pi}$, mm
e	Thermal effusivity, $\sqrt{k\rho C_p}$, $W \cdot s^{1/2}/(m^2 \cdot K)$
g	Acceleration of gravity, $m \cdot s^{-2}$
h	Droplet height, m
k	Thermal conductivity, W/(m·K)
q	Heat flux, W/m^2
Re	Reynolds number, $\rho D_0 V_0/\mu$
T	Temperature, ℃
T_d	Initial temperature of the droplet, ℃
T_f	Freezing temperature of water, ℃
T_s	Substrate temperature, ℃
t	Time, s
V_0	Impact velocity, $\sqrt{2gh}$, m/s
We	Weber number, $\rho D_0 V_0^2/\sigma$

Greek Symbols

α	Thermal diffusivity, $k/\rho C_p$, m^2/s
δ	Thickness, mm
ΔT_s	Substrate supercooling, T_s-T_f, ℃
μ	Dynamic viscosity, kg/(m·s)
ρ	Density, kg/m^3
σ	Surface tension, N/m
τ_s	Maximum spreading time, s

Subscripts

d	Droplet
f	Freezing
int	Interface
m	Maximum
l	Liquid water
s	Substrate
th	Thermal boundary layer

Abbreviations

fps	Frame per second
COD	Coefficient of determination

for the viscous and capillary regimes and the regime between the them, also at room temperature. For the cases with cold surface, during the drop spreading, a thin layer of ice can develop between the water and the substrate and pins the contact line at its edge when the drop reaches its maximal diameter. Thiévenaz et al. [42] divided the droplet impact freezing stage into the freezing of "pancake" and the liquid film above it which ultimately help characterize the thickness of the ice pancake. Fang et al. [43] further demonstrated that the freezing behavior of the water film above the "pancake" determines the freezing morphology. In addition, icing dynamics including spreading, freezing morphology, and freezing time, both on horizontal and inclined surfaces, on hydrophilic and hydrophobic surfaces, and on ice and non-ice surfaces, have been investigated comprehensively [44–52]. Jin et al. [45] observed the pinning of the droplet's contact line on the ice without recoiling upon reaching maximum contact diameter. The study also notes that lower ice surface temperatures lead to reduced spreading factors and increased ice bead heights, while higher droplet heights result in greater spreading and shorter freezing times, regardless of ice temperature. Thiévenaz et al. [47] verified that the final shape of the frozen drop is determined by the competition between the dynamics of retraction and freezing. Particularly, the water film retracts at a constant velocity which does not depend on the temperature nor on the film thickness. Zhu et al. [48] focused on the frozen morphologies of droplets impacting an inclined superhydrophilic surface under various substrate temperatures and impact velocities. The experiments reveal four new frozen morphologies: elliptical cap, half ring + cap I, half ring + cap II, and half ring + single ring. The study shows that these morphologies can be controlled by adjusting substrate temperature and impact velocity.

In recent years, Ghabache et al. [53], Ruiter et al. [54], Kant et al. [55] and Chen et al. [56], though with various liquids, have observed the interesting solidification, cracking, mechanical and morphological behaviors with the intricate interplay of fluid dynamics, heat transfer, and phase change with large substrate supercooling. For water droplets, Song et al. [57], Jin et al. [58] and Fang et al. [59] have also reported the self-peeling and ultra-low adhesion during droplet impact icing and well characterized the intrinsic physical mechanisms, which broaden the understandings on the fast freezing dynamics at interfaces. The interesting phenomena mainly lie in the interfacial fast freezing and subsequent thermal mechanical force relaxation and the first one plays as a premise of the second one. The instantaneous interface solidification upon contact will fundamentally influence the interactions between compressed air, water vapor, fog, frost, and ice [60], hence the subsequent macroscopic interfacial freezing, droplet spreading, icing dynamics and freezing time. In spite of the successful visualization of the air film between the impacting droplet and solid substrate [61–63], the effects of interfacial quick freezing on the subsequent spreading dynamics and freezing behavior are still not clear [58,64].

To elucidate the relationship between interfacial instantaneous freezing and the icing dynamics, in this study, the interfacial boundary layer freezing is accelerated by increasing the substrate supercooling and thermal effusivity. Then the interfacial freezing is characterized by the high-speed visualization. The abnormal icing dynamics including the spreading diameter and freezing time are well explained by the interfacial freezing phenomena and the heat transfer analysis. The present investigation unveils the mechanisms behind the fast interfacial freezing induced icing dynamics and can serve as the guidance for the design of icephobic surfaces in polar region or outer space at extreme low temperatures.

The manuscript is organized as follows. Section 2 introduces the experimental setup and the methods to extract data. Section 3 shows the icing dynamics phenomena, demonstrates the mechanisms of the abnormal spreading dynamics and presents the icing time and a revised model to better predict the icing time of the cases with high impact velocities and low substrate subcooling. Finally, several conclusions are drawn in Section 4.

2. Experimental methods

2.1. Test setup

The pictures of the experimental setup are shown in Fig. 1 which is primarily consisted by an acrylic box, a droplet generator (syringe pump and needle), cooling unit including copper cooling stage, test sample and refrigerator, nitrogen supply, temperature measurement and high speed imaging system. The temperature of the sample and the droplet impact height can be adjusted during the test. Before and during the experiment, high purity nitrogen (99.9 %) is discharged into the acrylic box to inhibit the frosting as much as possible. The main parameters influencing the droplet impact dynamics in this study include the droplet diameter D_0, impact velocity V_0, and substrate supercooling ΔT_s ($\Delta T_s = T_f - T_s$, in which T_s is the temperature of substrate and $T_f = 0$ ℃ is the freezing temperature of water at standard atmospheric pressure). High

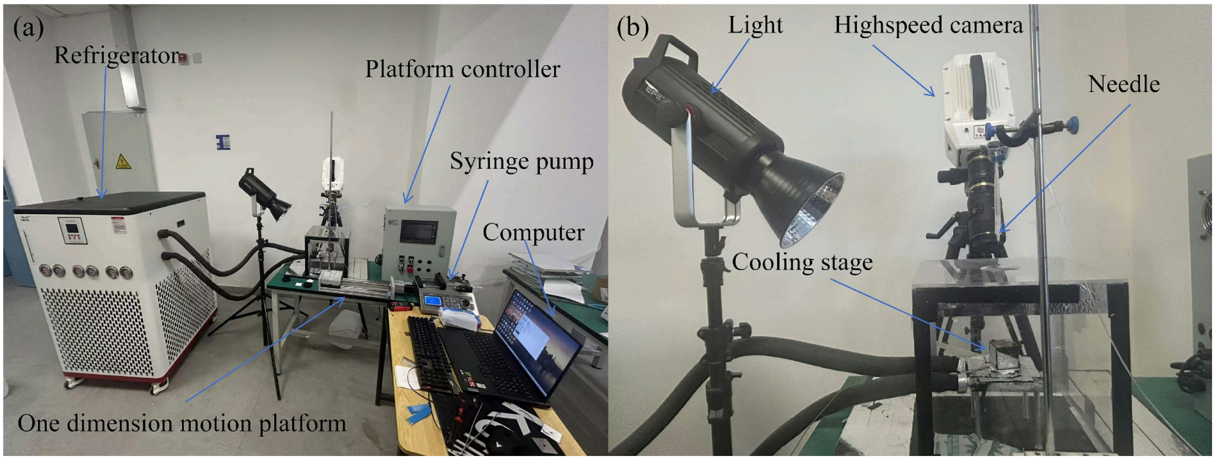

Fig. 1. Experimental setup (a) full view and (b) partial view.

speed camera (Revealer X113M) is used to capture the icing dynamics at a frame rate of 6800 fps and a resolution of 1280 × 1024 pixels. The droplet diameter is calibrated with a standard needle and keeps at 2.5 mm for all tests. The maximum spreading diameter is measured through the images captured by a cellphone camera from top view which will be elaborated below. The freezing time is also recorded by the high speed camera. For the precise control of the droplet impact site, an one dimension motion platform is used to adjust the position of the needle. To accelerate the interfacial freezing, bare copper plates with dimension of 40 × 25 × 1 mm are placed on the cooling stage as the test samples and the static contact angle between the bare copper and water is around 65°.

2.2. Data reduction

The spreading and freezing diameter of droplet impact icing are sometimes different and sometimes identical which is determined by whether the three phase contact line at maximum spreading is pinned at the maximum spreading position. In this study, since the substrate supercooling is high enough, all the impact droplets are pinned at the maximum spreading position and freeze. Thus, the spreading and freezing diameters of droplet are identical within this study.

Since the liquid film caused by the impacting droplet has a certain thickness, the nominal maximum spreading diameter, which has been widely used in many studies [14,18,41,44,48], is actually slightly larger than the real spreading diameter at the interface, as shown in Fig. 2. The real spreading diameter at the substrate-droplet interface can be further interpreted as the icing diameter when the liquid film has been pinned. Thus, we claim that in this study, the real maximum spreading diameter is measured and used for the following analysis.

For simplicity of the measurement, the maximum spreading diameter is measured after bulk freezing. In addition, we visualized the interfacial icing from the bottom of a sapphire sample, as shown in Fig. 3(a). The image shows that the ice contour is sometimes irregular, not a perfect circle so that the measurement from side-view will cause some error, which also supports the data reduction method used in this study. In this case, it needs a precise image processing to distinguish the exact spreading area which is governed by the inertia force, viscous force and surface tension. As shown in Figs. 3(b) and (c), by incorporating this method in the ImageJ software, the spreading area A_m can be precisely measured. Then the maximum spreading diameter D_m can be estimated by $\sqrt{4A_m/\pi}$ by assuming the spreading area as a circle.

3. Results and analysis

3.1. Icing dynamics

To study the icing dynamics with instantaneous interfacial freezing during droplet impact icing, tests were conducted with substrate supercooling from 29 ℃ to 64 ℃ and impact velocity from 1.88 m/s to 3.13 m/s. Fig. 4 presents the time-lapse images of the droplet's impact, spreading and freezing dynamics under four typical working conditions. Specifically, as shown in Figs. 4(a) and (b), at a low ΔT_s of 29 ℃, when the droplet expanded to its maximum size, it took on a sunflower-like pattern. When the droplet completely froze, cracks and a cap formed on the top when the impact velocity is as low as 1.88 m/s. In the meanwhile, spreading and freezing time were shortened by a much higher impact velocity of 3.13 m/s and there was no cap but a circular ring instead. As the ΔT_s increased to 64 ℃, the spreading time was shortened due to the more rapid cooling for both cases in Figs. 4(c) and (d). As to the spreading morphology, the droplets were spread into a relatively standard circular shape and splashing shape, respectively. However, the maximum spreading time was prolonged with increased impact velocity, not as that of $\Delta T_s = 29$ ℃. Another phenomenon caused

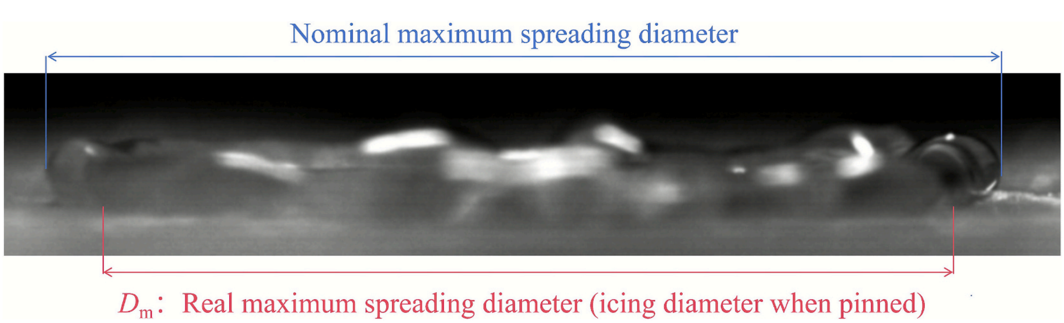

Fig. 2. Comparison between the nominal maximum spreading diameter with the real maximum spreading diameter.

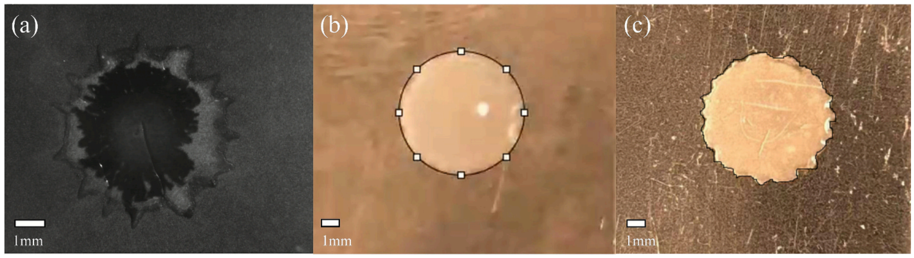

Fig. 3. (a) Interfacial icing area after droplet impact from the bottom view, $\Delta T_s = 50℃$, $V_0 = 2.62$ m/s and method used to measure the droplet spreading area by ImageJ with (b) $\Delta T_s = 50℃$, $V_0 = 1.88$ m/s and (c) $\Delta T_s = 36℃$, $V_0 = 1.88$ m/s.

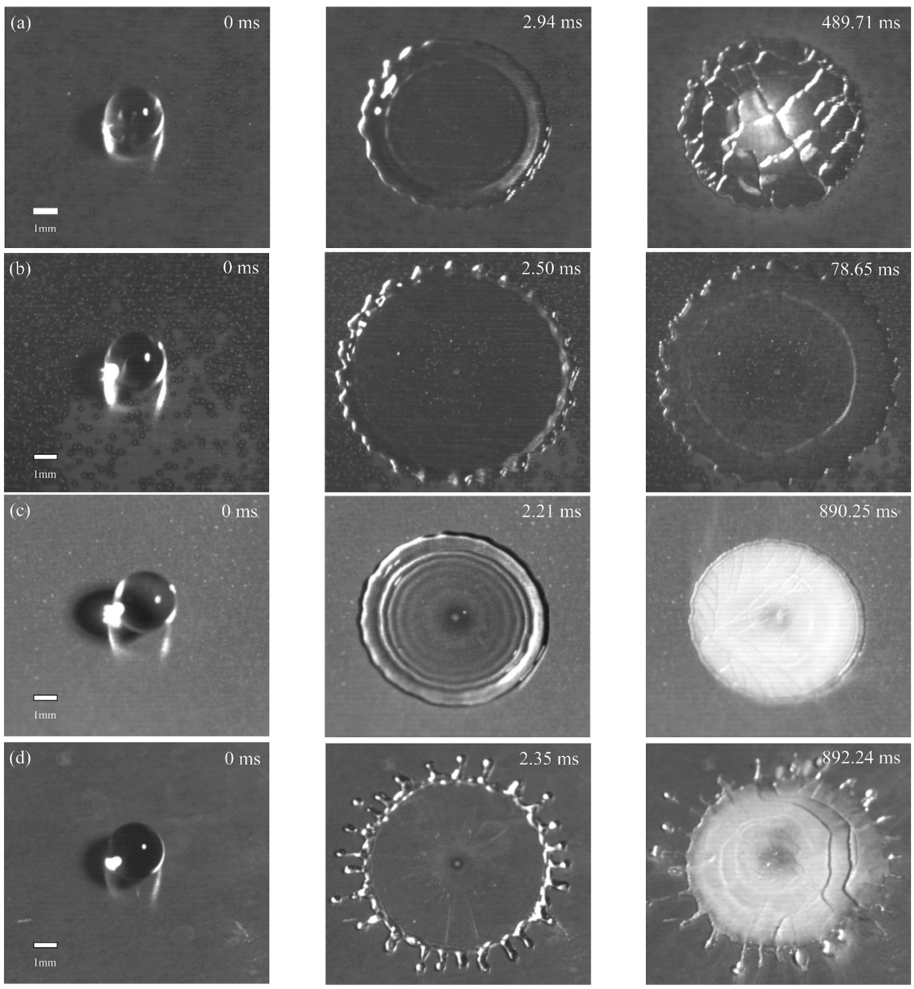

Fig. 4. Icing dynamics under the conditions of (a) $\Delta T_s = 29℃$, $V_0 = 1.88$ m/s, (b) $\Delta T_s = 29℃$, $V_0 = 3.13$ m/s, (c) $\Delta T_s = 64℃$, $V_0 = 1.88$ m/s and (d) $\Delta T_s = 64℃$, $V_0 = 3.13$ m/s.

by the increase ΔT_s lies in the formation of frost between ice lamella and substrate manifested by the white color in the last image of Figs. 4(c) and (d), as observed by Jin et al. [55]. We also noticed that the frozen time increases instead of decreases at high ΔT_s, as shown in Figs. 4(c) and (d). An irregular water ring is faintly visible at the top of the droplet, and the complete freezing of the ring is a lengthy process. In addition, the droplet may burst apart at high impact velocity and high substrate supercooling due to the mechanical stress relaxation.

3.2. Spreading dynamics

Noticing the anomalous icing dynamics shown in Fig. 4. Tests were conducted under five impact velocities and six ΔT_s. As expected, the normalized maximum spreading D_m/D_0 decreases with increased ΔT_s due to the increased viscosity of water at droplet-substrate interface, as shown in Fig. 5. This variation trend has been reported by Fang et al. [43] and conforms to the classic droplet impact spreading theory where the increased viscous force will impede the spreading of impacting droplet [41]. However, the abnormal spreading behavior occurs at high ΔT_s especially over 57 ℃, the spreading diameter does not decrease further but even increases with further increases in ΔT_s, consistent with the findings in section 3.1 which means the droplet spreading under the conditions with quick interface freezing is also driven by other effects except inertial force, capillary force and viscous force. At the impact velocities of 3.13 m/s, 2.62 m/s and 2.21 m/s, the maximum spreading diameter of the droplet with $\Delta T_s = 64$ °C even surpasses the value

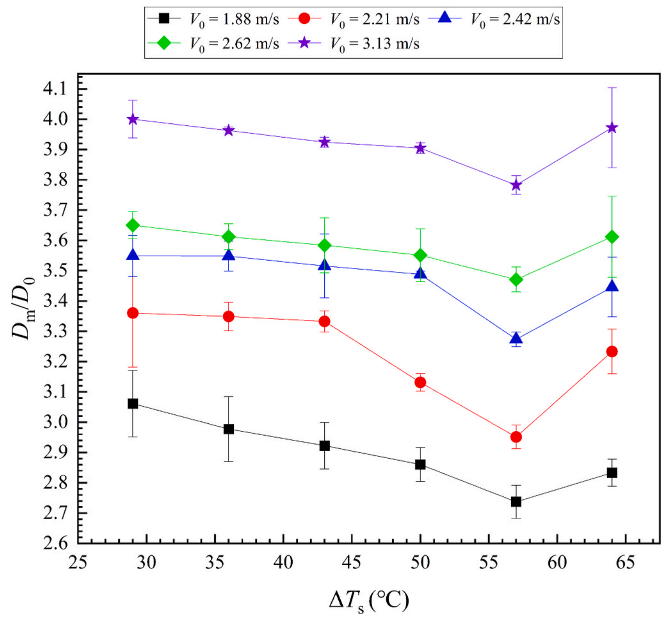

Fig. 5. The variation trend of dimensionless spreading diameter D_m/D_0 versus substrate supercooling ΔT_s.

observed at the same impact velocity with $\Delta T_s = 50$ °C.

To quantify the maximum spreading diameter under the condition of droplet impact icing with instantaneous interfacial freezing, we formulate the dimensionless spreading diameter with the formulation proposed by Laan et al. [41]. The capillary force and viscous force in this case are quantified in terms of Weber number (We $= \rho D_0 V_0^2/\sigma$) and Reynolds number (Re $= \rho D_0 V_0/\mu$), where ρ is the density, σ is the surface tension and μ is the dynamic viscosity of the water. Since the surface tension and viscosity of water are temperature-dependent, the characteristic temperature of the droplet during impact can be written as Eq. (1) [43].

$$T = T_d - \int_0^{\tau_s} q dt/(\rho C_p \delta_{th}) \tag{1}$$

In which $T_d = 23$ °C is the initial temperature of the droplet, $\delta_{th} = 2.8\sqrt{\alpha_l \tau_s}$ refers to the thickness of thermal boundary layer while τ_s is defined as the time taken for the droplet expands to its maximum diameter, α_l and C_p refer to the diffusivity and specific heat of the water respectively and q is the heat flux during impacting and it can be written as Eq. (2) [40].

It should be noted that diffusivity α can be calculated as $\alpha = k/\rho C_p$.

$$q = e_l e_s (T_d - T_s)/[(e_l + 0.79 e_s)\sqrt{\pi}\sqrt{t}] \tag{2}$$

Where e_l and e_s refer to the effusivity of the water and substrate respectively and t is the time, in which effusivity e can be calculated as $e = \sqrt{k\rho C_p}$. Based on this, we can obtain the physical properties of supercooled water at T. Then, the density can be derived by extracting the data provided by Holten et al. [65], and the surface tension can be calculated as $\sigma = 235.8(1\text{-}T/647.096)^{1.256}[1\text{–}0.625(1\text{-}T/647.096)]$ [66]. We approximated the viscosity of supercooled water by fitting the viscosity curve of water from the NIST database [67], which can be written as $\mu = 0.079\text{–}5.708 \times 10^{-4}T + 8.252 \times 10^{-7}T^2$, whose COD (Coefficient of determination) R^2 reaches 0.999. The rescaled maximum spreading ratio can be plotted as a function of WeRe$^{-2/5}$, both dimensionless parameter We and Re can be calculated with the characteristic temperature expressed by Eq. (1). The complete expression can be written as Eq. (3) [41].

$$(D_m/D_0)Re^{-1/5} = \left(We\,Re^{-2/5}\right)^{1/2}/\left[A + \left(WeRe^{-2/5}\right)^{1/2}\right] \tag{3}$$

According to the classical spreading theory, the lower substrate temperature means the larger viscosity of the liquid water which impedes the spreading of droplet. Therefore, we removed the five sets of data with $\Delta T_s = 64$°C in Fig. 5 that does not conform to the theoretical trend when we performed the fitting. The fitting results based on Eq. (3) are shown in Fig. 6 and the coefficient A is fitted to be 1.62 ± 0.03 and the $R^2 \approx 0.86$.

To further investigate the interfacial effects, we captured snapshots of the droplets at their maximum spreading diameter under various working conditions, as shown in Fig. 7. We observed that at lower ΔT_s (29 °C – 36 °C), the droplet-substrate contact is relatively tight, demonstrated by the homogeneous dark gray or black color. And in these cases, a tiny ice dot forms at the initial contact point, as shown by the images in black box. In contrast, at higher ΔT_s (43 °C – 64 °C), a smaller, closely packed contact area (indicated by black ring) appears around the ice dot, surrounded by frost layer (white area), as indicated by the images in blue box. This frost area increases with higher substrate supercooling and impact velocity. When the impact velocity or supercooling degree is sufficiently high, the frost layer exhibits a splashing pattern, demonstrated by the images in red box.

To elucidate the phenomena observed in Fig. 7, we provide schematic illustrations of the impact-spread-freeze stages of droplet impacting to a large supercooled surface respectively, as depicted in Fig. 8. Before impacting the substrate (Fig. 8(a)), the surface of the droplet is already surrounded by water vapor. After the droplet impacts the upper surface of substrate (Fig. 8(b)), the central region consists of an ice dot and a closely adhered ice ring that is in contact with the substrate surface. At the same time, the compressed air between the droplet and substrate is flowing outward from all directions carrying water vapor which induces a large amount of frost forms between the droplet and upper surface of substrate with circle or splashing patterns, as indicated by Fig. 7 and Fig. 8(b). Obviously, the faster the impact velocity, the faster the diffusion of compressed air. Moreover, when the degree of supercooling is higher, the bottom of the droplet is more likely to freeze. Therefore, when ΔT_s and the velocity are high enough, a splashing frost layer will more likely to form at the bottom of the droplet. After the droplet completely frozen (Fig. 8(c)), frost still exists, as manifested by the white color within the droplet impact region in Figs. 4(c) and (d).

In addition, we speculate that the flowing compressed air also serves as a lubrication layer between droplet and substrate during spreading, similar like the drag reduction effect of air film created by superhydrophobic surfaces [68–70] and Leidenfrost phenomena [71]. The lubrication effect accounts for the abnormal spreading behavior shown

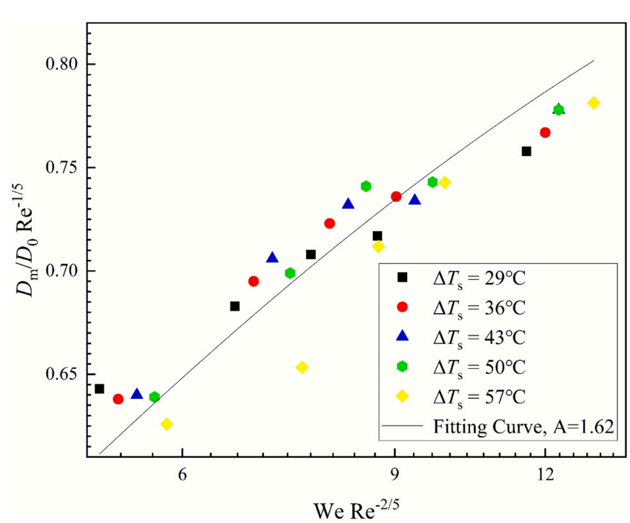

Fig. 6. The fitting curve of droplet normalized maximum spreading ratio (D_m/D_0) Re$^{-1/5}$ as a function of WeRe$^{-2/5}$.

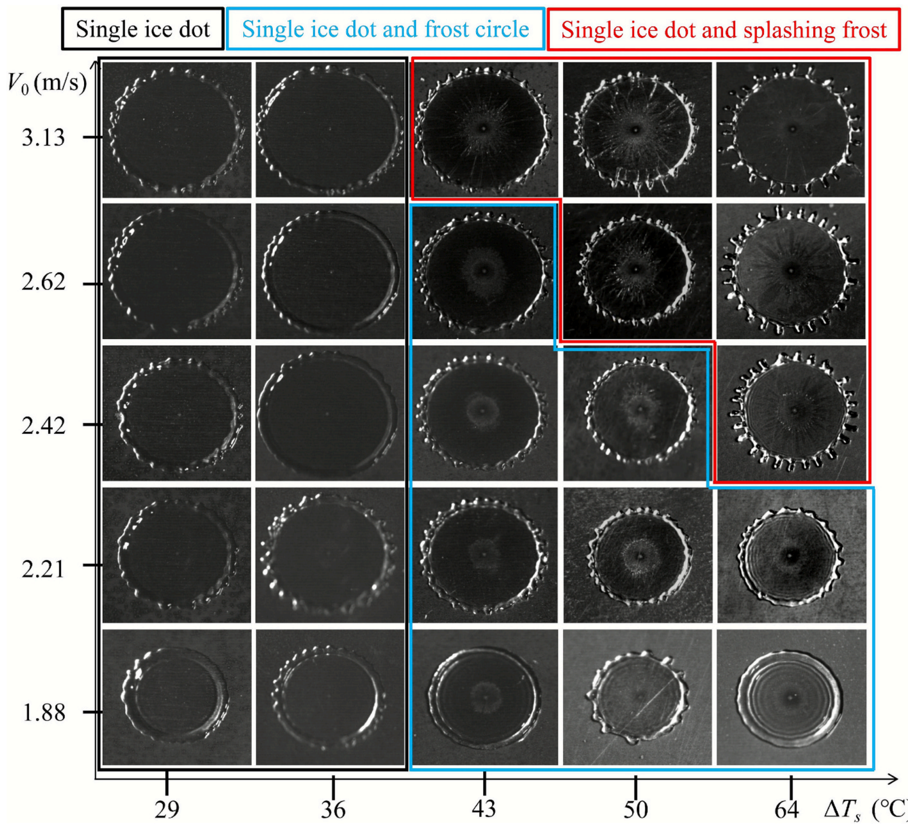

Fig. 7. Instantaneous snapshots of droplets under various operating conditions when expand to maximum diameter.

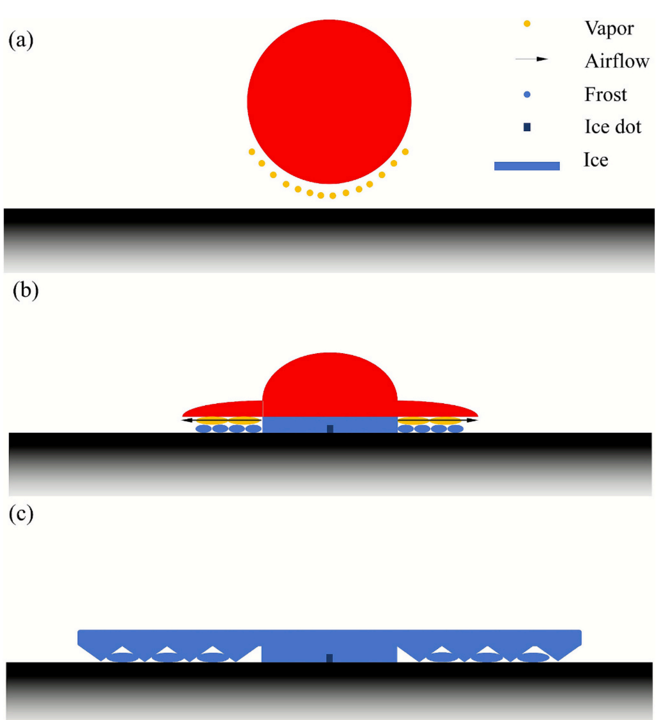

Fig. 8. Schematic of instantaneous interfacial freezing during droplet impact onto a large supercooling substrate at high impact velocity (a) Before impact, (b) Spreading stage, (c) After freezing.

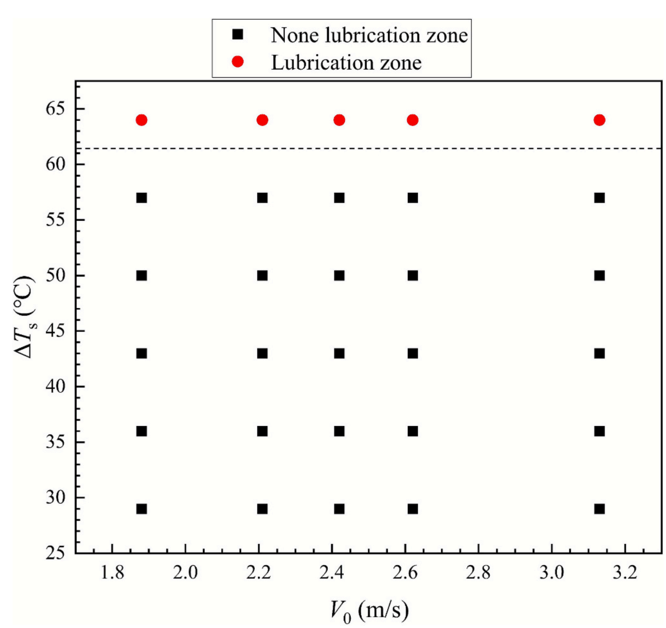

Fig. 9. Phase diagram of maximum spreading during droplet impact icing.

in Fig. 5. As shown in Fig. 9, we have defined a lubrication zone, where the cases with an increase in D_m at larger substrate supercooling belongs to this zone, and the lubrication zone generally corresponds well with with presence of splashing frost on the interface in Fig. 7.

3.3. Freezing time

The freezing time under various working conditions is shown in

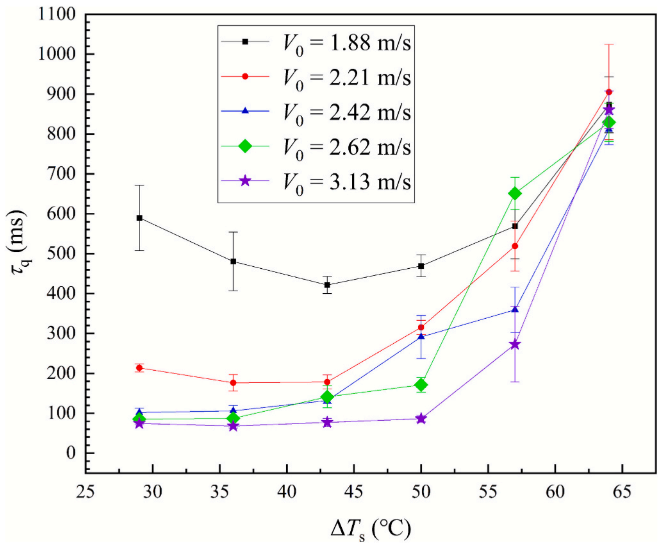

Fig. 10. Bulk freezing time versus substrate supercooling under five impact velocities.

Fig. 10. At low ΔT_s(≤43℃) and impact velocities (≤2.21 m/s), the freezing time decreases with the increase of ΔT_s. However, when the impact velocity reaches a certain level (≥ 2.42 m/s) and $\Delta T_s \le 43$℃, the freezing time slowly increases. Moreover, when ΔT_s reaches a certain level (43℃), the freezing time increases sharply for all cases, with the timescale extending from hundreds of milliseconds to nearly one second. The prolonged freezing duration is attributable to the poor contact between droplet and substrate, as reported by Jin et al. [55].

Currently, there are several models available for predicting the freezing time of impacted droplets. To achieve more accurate predictions, calculating the interfacial temperature between droplet and substrate is crucial Ghabache et al. [53] approximated the interfacial temperature as being equal to the substrate temperature. However, Ruiter et al. [54] demonstrated through calculations that the latent heat of the freezing process must be taken into account to accurately determine the interfacial temperature. In light of the heat absorbed during the solidification of the droplet, a more accurate method for calculating the interfacial temperature has been incorporated to revised the freezing time prediction model. It can be written as Eq. (4), named as Schwartz model [72]:

$$T_{\mathrm{d,int^*}} = T_s + \frac{k_f \alpha_s^{1/2} \Delta T_s}{k_f \alpha_s^{1/2} + k_s \alpha_f^{1/2} erf(\lambda)} \tag{4}$$

where k_f and k_s refer to the thermal conductivity of the ice and substrate respectively and α_f refers to the diffusivity of ice. And λ is a constant, whose numerical solution can be calculated from Eq. (5):

$$\frac{k_s \alpha_f^{1/2} e^{-\lambda^2}}{k_f \alpha_s^{1/2} + k_s \alpha_f^{1/2} erf(\lambda)} - \frac{k_l \alpha_f^{1/2} (T_d - T_s) e^{-\alpha_f \lambda^2/\alpha_l}}{k_f \alpha_l^{1/2} \Delta T_s erfc[\lambda(\alpha_f/\alpha_l)^{1/2}]} = \frac{\lambda L \pi^{1/2}}{C_{p_f} \Delta T_s} \tag{5}$$

where k_l and α_l refer to the thermal conductivity and diffusivity of water respectively and L refers to the latent heat of water. Note that

$$erf(x) = \frac{2}{\sqrt{\pi}} \int_o^x e^{-y^2} dy \tag{6}$$

$$erfc(x) = 1 - erf(x) \tag{7}$$

Fig. 11 presents a comparison of the interfacial temperature calculated using the Schwarz model by Eq. (4) with Ts and those obtained from the conduction model where the latent heat of water during freezing is neglected [72]. As can be observed, the interfacial temperature derived from the Schwarz model are indeed higher since the latent

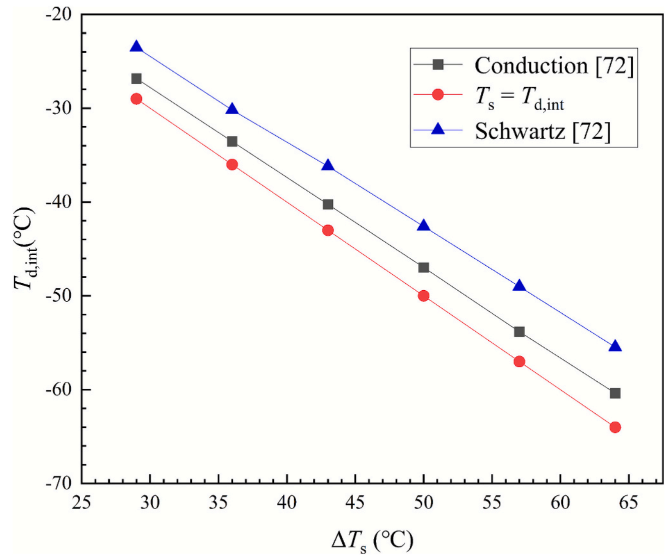

Fig. 11. Comparison among T_s, interfacial temperature calculated from conduction model ($T_{d,int}$) and Schwartz model(T_{d,int^*}) [72].

heat is considered in the calculation of interfacial temperature, as shown in Eq. (4).

Then the bulk freezing time of the impact droplet can be estimated by Eq. (8) which is revised, by substituting the interfacial temperature with Eq. (4), based on the ice pancake heat transfer model used by Ghabache et al. [53]:

$$\tau_{bf} = \frac{\rho_f L h_0^2}{K_f (T_f - T_{d,int^*})} \tag{8}$$

where h_0 can be calculated as $2D_0^3/3Dm^2$.

Then we compare the predicted freezing time obtained by Eq. (8) with revised interfacial temperature (Eq. (4)) with experimental data in Figs. 12 and 13. The predicted data fits the experimental data comparatively well with relative deviation between −24.29 % and 1.82 % for the 6 working conditions with $\Delta T_s \le 36$℃ and $V_0 \ge 2.42$ m/s as shown in Fig. 12. However, as shown in Fig. 13, the discrepancy is quite significant for the other 24 cases with relative deviations range from

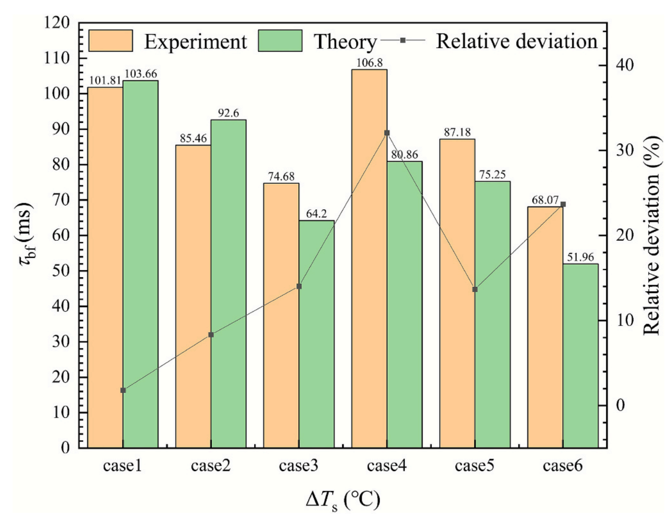

Fig. 12. Comparison of freezing time between the experimental data and the predicted value: Case 1 with $\Delta T_s = 29$℃ and $V_0 = 2.42$ m/s, Case 2 with $\Delta T_s = 29$℃ and $V_0 = 2.62$ m/s, Case 3 with $\Delta T_s = 29$℃ and $V_0 = 3.13$ m/s, Case 4 with $\Delta T_s = 36$℃ and $V_0 = 2.42$ m/s, Case 5 with $\Delta T_s = 36$℃ and $V_0 = 2.62$ m/s and Case 6 with $\Delta T_s = 36$℃ and $V_0 = 3.13$ m/s.

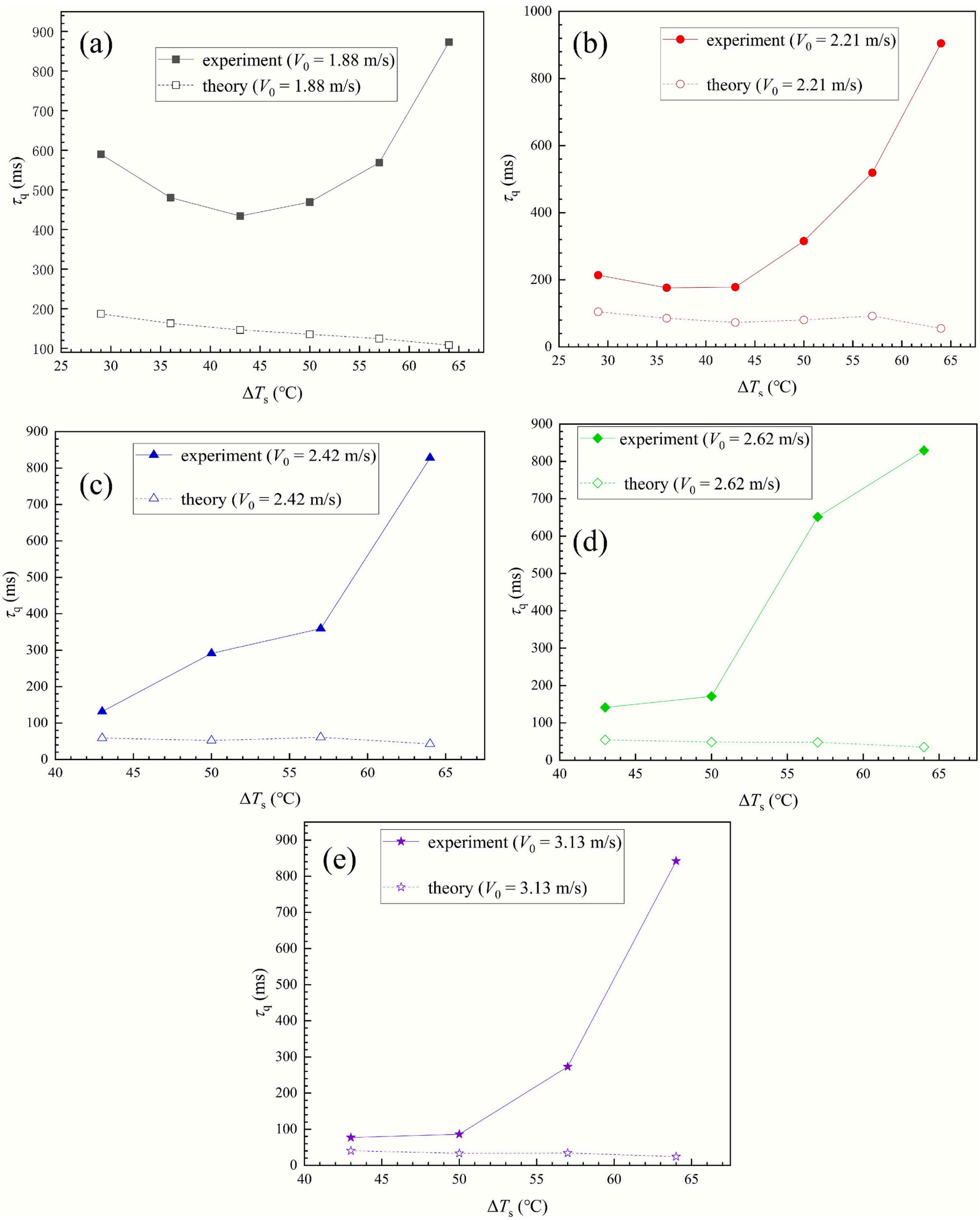

Fig. 13. Comparison of freezing time between the experimental data and the predicted value by Eq. (8) of cases with poor fit: (a) $V_0 = 1.88$ m/s, (b) $V_0 = 2.21$ m/s, (c) $V_0 = 2.42$ m/s, (d) $V_0 = 2.62$ m/s, and (e) $V_0 = 3.13$ m/s.

−93.89 % to −47.93 %.

This is because, for the well fitted cases, the substrate supercooling is under 43℃, which falls into the sticking regime proposed in Ref. [58] which ensures a good contact between droplet and substrate. In addition, from the morphology observation in Figs. 4(a) and (b), high impact velocity makes the ice lamella more close to a pancake shape and enhances the heat transfer between droplet and substrate. Thus, heat transfer process of the six cases with low ΔT_s and large V_0 can be well formulated by the revised model (Eq. (8)) in this study. On the contrary, the model performs poorly in predicting the freezing time of "bad contact" cases with a huge thermal resistance caused by the air film between ice and substrate.

4. Conclusions

This study investigates the abnormal icing dynamics induced by instantaneous interfacial freezing during droplet impact icing by increasing the substrate supercooling and thermal conductivity, the summary is as follows.

1) A new maximum spreading diameter measurement method is proposed to better characterize the interfacial spreading dynamics during impact icing with quick interface freezing.
2) When the substrate supercooling exceeds a certain threshold value (57℃), the spreading diameter exhibits an increasing trend with further increase in substrate supercooling, contradicting with the classical impact spreading theory.
3) Three interfacial freezing morphologies, including single ice dot, single ice dot and frost circle and single ice dot and splashing frost, during droplet spreading are indentified with varied substrate supercooling and impact velocity. The splashing frost is likely to form with increased ΔT_s and V_0 due to the fast escaping of compressed air outward.
4) The freezing time of droplet impact icing also exhibits abnormal variation trends with varied substrate supercooling and impact velocity, especially under conditions of bad contact with $\Delta T_s \geq 43℃$ where freezing time increases sharply with increased ΔT_s.
5) A revised freezing time prediction method is proposed by introducing the interface temperature, taking into the latent heat into consideration during solidification, and fits well with the experimental data of the cases with good contact between droplet and substrate.

The present study deepens the understanding of mechanisms behind the abnormal icing dynamics induced by instantaneous interfacial freezing and paves the way to develop more accurate heat transfer models during droplet impact icing. The unprecendented insights highlight the need for further in-depth research and more advanced prediction models accounting for interfacial effects on icing behaviors.

CRediT authorship contribution statement

Puhang Jin: Writing – original draft, Visualization, Validation, Resources, Methodology, Investigation, Funding acquisition, Conceptualization. **Yuqiao Chen:** Writing – original draft, Visualization, Validation, Methodology, Investigation, Data curation. **Jiale Sun:** Writing – review & editing, Visualization, Methodology, Investigation. **Zhiming Tan:** Writing – review & editing, Visualization, Methodology, Investigation. **Wanqing Zhao:** Validation, Investigation. **Gongnan Xie:** Writing – review & editing, Supervision, Resources, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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